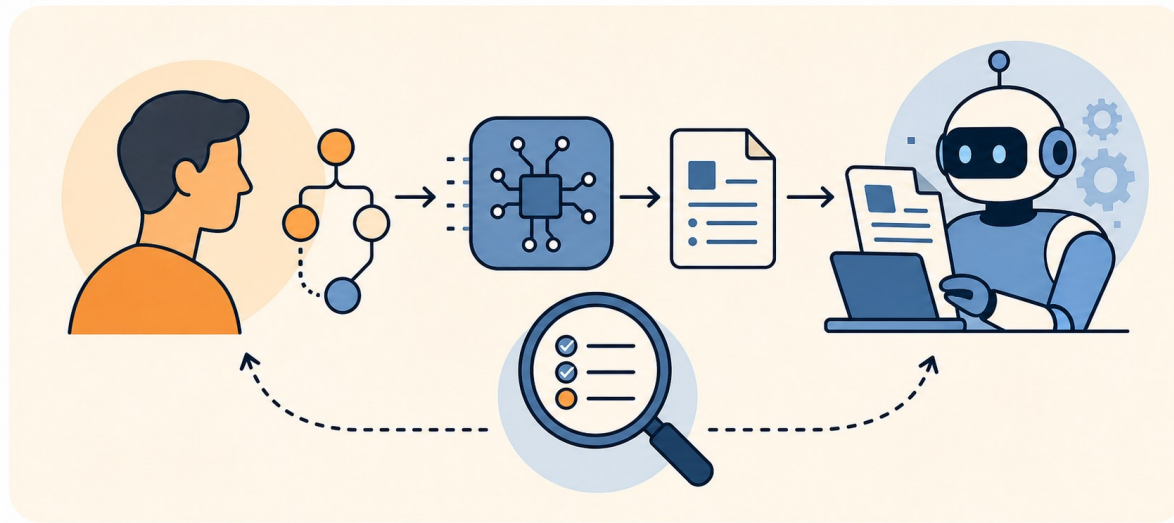


Delegation and Verification Under AI

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How does AI reshape worker quality in the AI Age?

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Paper : <https://arxiv.org/abs/2603.02961>

The Puzzle: Cheaper Execution even Lowers Worker Quality

Expected effect

AI assistance



Lower execution cost

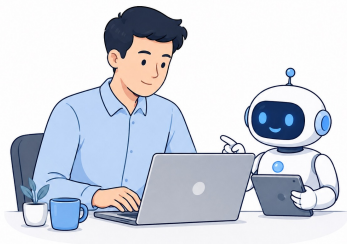
Higher execution efficiency



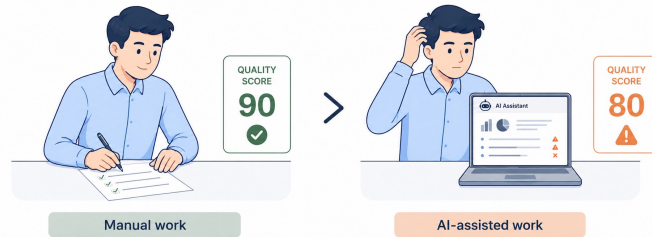
Always lead to better worker quality ?



Better AI, larger quality improvement ?



Observed puzzle



- **(Over-reliance)** Workers rely more on AI advice on harder tasks with lower AI accuracy [Zhang & Deniz, 2024]
- **(Oversight gap)** High-load tasks reduce oversight when AI errors need more scrutiny [Gaube et al., 2021]
- **(Mixed impact)** AI upgrades some workers but downgrades others, despite similar AI access [Brynjolfsson et al., 2025]

Existing explanations

- Attribute oversight decay to cognitive load, verification complexity, and time pressure. [Bergman, 2025]
- Treat uneven AI effects as task allocation or fixed human-skill differences. [Celis et al., 2025]

What is missing?

Underlying mechanism on how AI reshape institutional worker quality is still unclear

Our Contributions

- **(Rational action)** Model worker action (delegation & verification) as optimality for individual utility (task success v.s. cost)
- **(Worker quality)** Define institutional worker quality as institutional gains induced by worker's optimal action
- **(Compliance regime)** Characterize when AI assistance improves or degrades institutional worker quality; identify which workers are upgraded or downgraded
- **(Verification amplifier)** Obtain closed-form expressions for optimal actions and resulting worker quality, which highlights the importance of verification reliability

Cheaper execution helps only when workers can verify

Model: Delegation, Verification, and Worker Quality

Action space & workflow

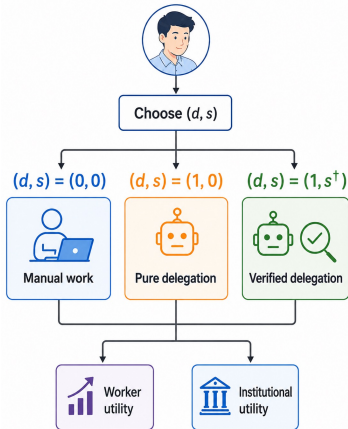
We study a static, per-task model of AI-assisted work

Worker action space:

- $d \in [0,1]$: delegation prob.
- $s \in [0,1]$: verification effort after delegation

Workflow logic:

- **Manual work**: worker performs the task **manually**
- **Pure Delegation**: worker takes AI output directly
- **Verified delegation**: worker **checks** AI output



Worker utility

- $p(d, s)$: the success probability
- $cost(d, s)$: the incurred cost
- b_W/ℓ_W : worker's success gain/failure loss

Worker utility: $U_W(d, s) := b_W p(d, s) - \ell_W(1 - p(d, s)) - cost(d, s)$

Worker action -- **Optimizing worker utility**
 $(d^*, s^*) \in \arg \max_{(d,s) \in [0,1]^2} U_W(d, s).$

Worker quality

- b_I/ℓ_I : institution's success gain/failure loss
- ξ : institution-paid worker cost share
- Assume $b_I + \ell_I > \xi(b_W + \ell_W)$, e.g., $b_I = \ell_I, b_W = \ell_W, \xi = 0.3$

Institutional utility: $U_I(d, s) := b_I p(d, s) - \ell_I(1 - p(d, s)) - \xi cost(d, s)$

Institutional worker quality is induced by worker's **optimal action**: $Q(W) := U_I(d^*, s^*)$

- **Misalignment**: Since $U_I \neq U_W$, rational worker choice always improve worker utility, while can harm **institutional worker quality**!

Task success probability

Three task-success paths:

- **Manual work**: worker performs task, succeed with $(1 - d)p_w$, p_w : worker success rate
- **Pure AI success**: task delegated to AI, succeeds with dp_a , p_a : AI success rate
- **Detect & correct**: if AI fails, worker detects error with $\phi(s)$, redoes task, succeeds with p_w , $\phi(s)$: error-detection probability, e.g., $\phi(s) = 1 - e^{-\alpha s}$ with **verification reliability** α

Thus, $p(d, s) = (1 - d)p_w + dp_a + d(1 - p_a)\phi(s)p_w$

Cost function

Three cost components:

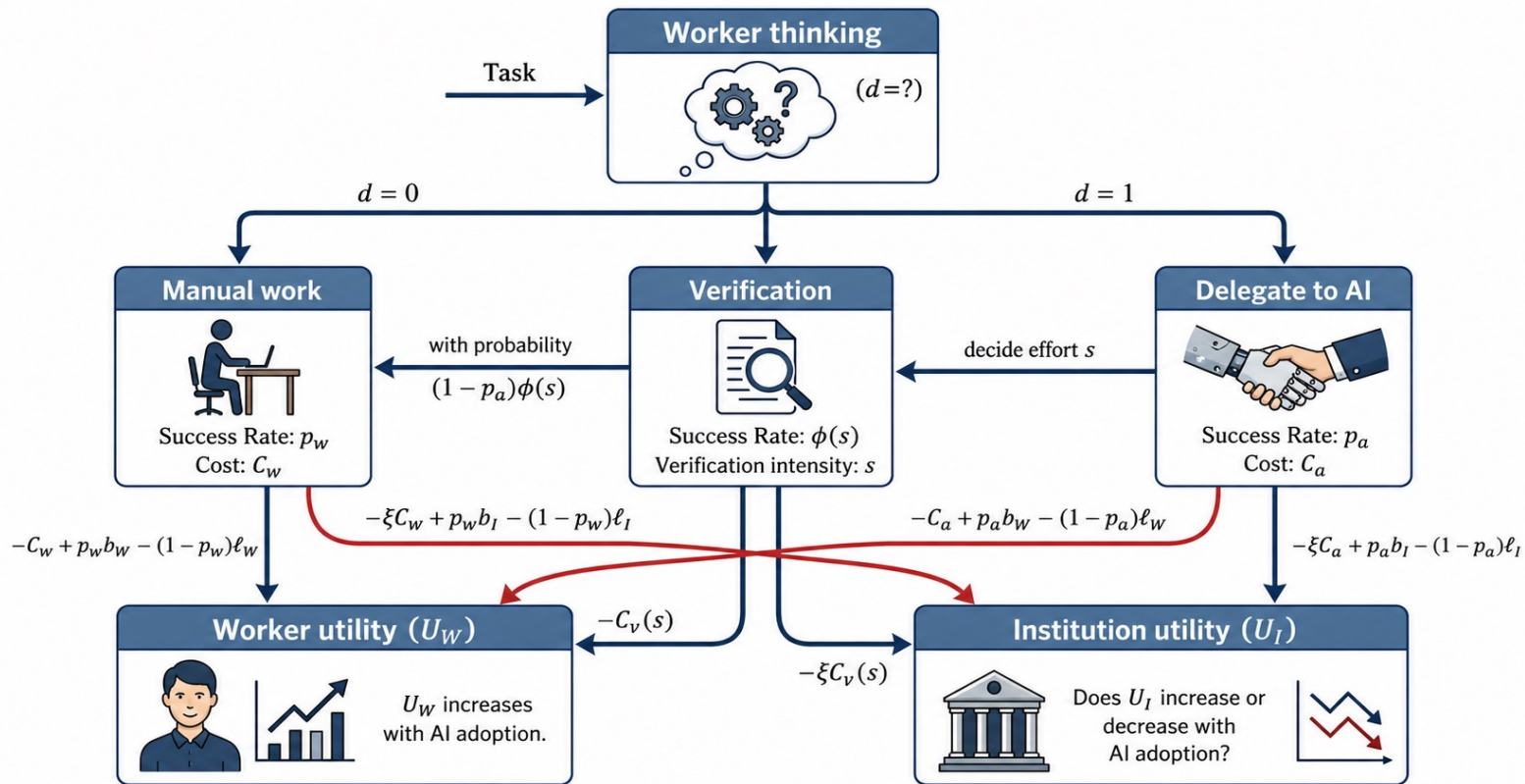
- **Manual work**: worker execution cost $(1 - d)C_w$, e.g., $C_w(\beta) = 1 - \beta$ with **execution efficiency** β
- **AI execution**: AI execution cost dC_a
- **Verify & correct**: worker verifies AI output at cost $C_v(s)$; if AI error detected, worker redoes task with cost C_w

$C_v(s)$: verification cost, e.g., $C_v(s) = s + \frac{s^2}{2}$

Thus, $cost(d, s) = (1 - d)C_w + dC_a + dC_v(s) + d(1 - p_a)\phi(s)C_w$

We analyze how optimal action (d^*, s^*) and quality $Q(W)$ vary with verification reliability α and execution efficiency β

Mechanism: How (d, s) Determines Worker Quality



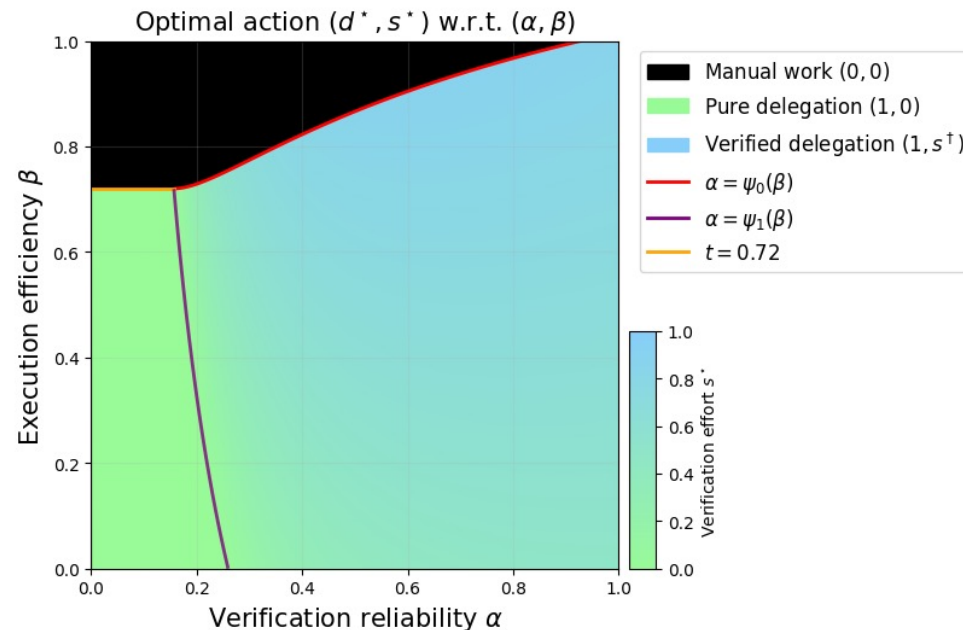
A flow chart for the workflow under AI assistance

Theoretical Result: Result 1 — Three Workflow Regimes

Theorem (Worker optimal action)

There exists threshold t , monotonically increasing function $\psi_0: [0,1] \rightarrow \mathbb{R}_{\geq 0}$, monotonically decreasing function $\psi_1: [0,1] \rightarrow \mathbb{R}_{\geq 0}$, function $s^\dagger: [0,1]^2 \rightarrow [0,1]$, such that the optimal action (d^*, s^*) takes the following form:

- **(Manual work)** $(0,0)$ if $\beta \geq t, \alpha < \psi_0(\beta)$
- **(Pure delegation)** $(1,0)$ if $\beta < t, \alpha < \psi_1(\beta)$
- **(Verified delegation)** $(1, s^\dagger(\alpha, \beta))$ otherwise ($0 < s^\dagger(\alpha, \beta) \leq 1$)



Analysis

• Three workflow regimes

Ability space (α, β) is partitioned into **manual work**, **pure delegation**, and **verified delegation**

• Sharp workflow switches

Small changes in (α, β) can **abruptly switch** workflow choices, consistent with observations in [Zhang & Deniz, 2024]

• Verification reliability drives oversight

When α is sufficiently high, **verification becomes worthwhile**, moving the worker from pure delegation or manual work to verified delegation

• Execution efficiency reduces AI reliance

Higher β lowers manual execution cost, making manual work more attractive and **reducing delegation incentives**

Verification reliability separates verified delegation from unchecked reliance.

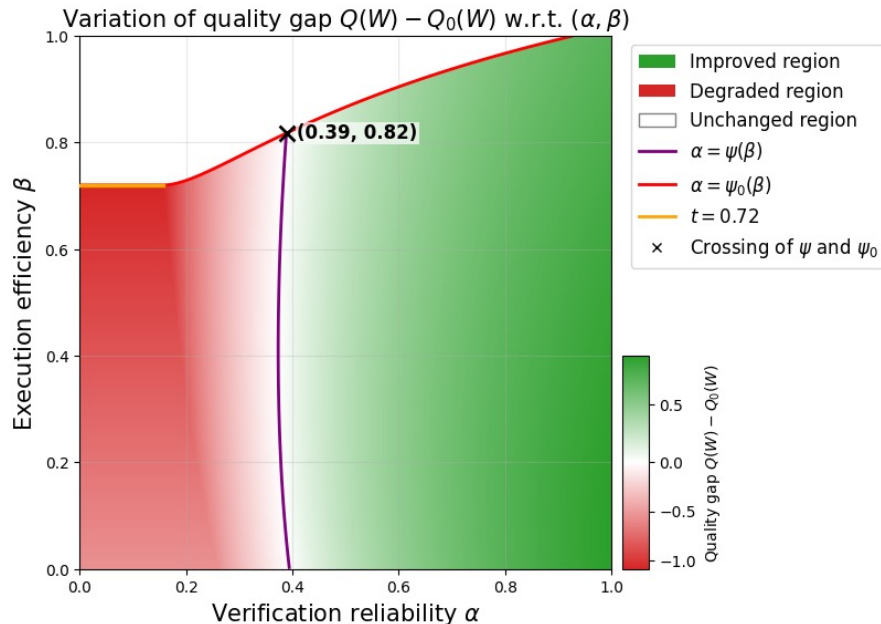
Theoretical Result: Result 2 — Quality Gap under AI

Theorem (Worker quality improvement v.s. no AI)

Let $Q_0(W) := U_I(0,0)$ denote no-AI baseline. There exists a continuous function $\psi: [0,1] \rightarrow \mathbb{R}_{\geq 0}$ such that:

- **(Improved)** $Q(W) > Q_0(W)$ if $\alpha > \psi(\beta)$
- **(Unchanged)** $Q(W) = Q_0(W)$ if $((\beta \geq t) \wedge (\alpha < \psi_0(\beta)))$ or $\alpha = \psi(\beta)$
- **(Degraded)** $Q(W) < Q_0(W)$ otherwise

Analysis



• Three quality regimes

Ability space (α, β) is partitioned into **quality gain, unchanged, and loss regimes**

• Execution efficiency alone is insufficient

Higher β improves no-AI baseline $Q_0(W)$, but AI's effect depends on induced workflow

• Quality loss stems from rational over-delegation

Worker-optimal delegation can be institutionally harmful, **without bias** or miscalibration

• Verification reliability separates gain from loss

High α enables error detection and correction. Low α leaves delegation **weakly supervised** and can reduce quality even under **rational** worker choice

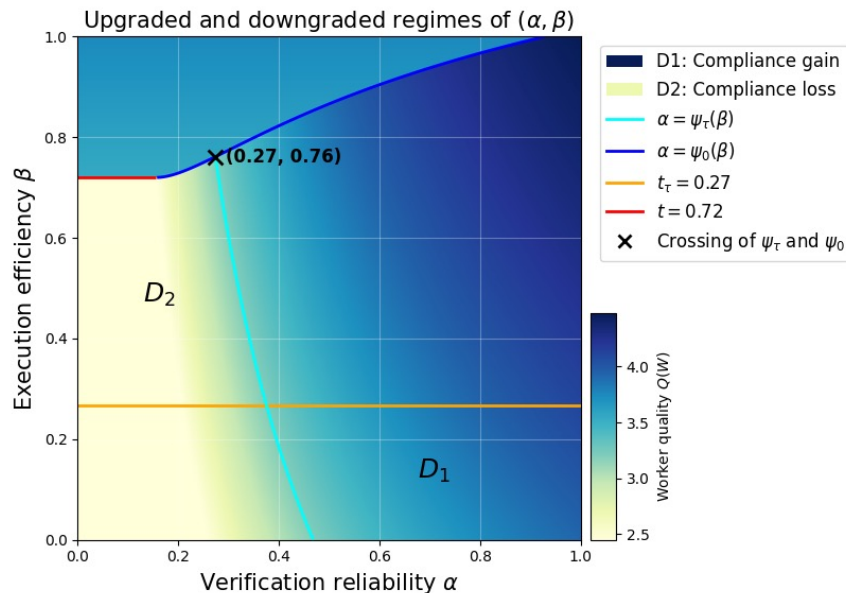
AI does not **uniformly** raise worker quality: **verification reliability α** determines whether delegation becomes a gain or a loss

Theoretical Result: Result 3 — Compliance Gain and Loss

Theorem (Worker upgraded v.s. downgraded)

Let τ be the institutional qualification threshold. There exists threshold t_τ , a decreasing boundary $\psi_\tau: [0,1] \rightarrow \mathbb{R}_{\geq 0}$ such that:

- Compliance gain ($(Q(W) > \tau \wedge Q_0(W) < \tau)$) holds iff:
 $(\beta < \min\{t, t_\tau\}) \wedge (\alpha > \psi_\tau(\beta))$ or $(t \leq \beta < t_\tau)$ or $\alpha > \max\{\psi_0(\beta), \psi_\tau(\beta)\}$
- Compliance loss ($(Q(W) < \tau \wedge Q_0(W) > \tau)$) holds iff:
 $(t_\tau < \beta \leq t) \wedge (\alpha < \psi_\tau(\beta))$ or $(\beta > \max\{t, t_\tau\}) \wedge (\psi_0(\beta) < \alpha < \psi_\tau(\beta))$



Analysis

- **High α enables compliance gain**

Workers **with low baseline** quality can become **qualified** when reliable verification makes AI delegation institutionally valuable

- **Low α causes compliance loss**

Workers who **were qualified** without AI may become **unqualified** through rational over-delegation and weak oversight

- **Verification amplification**

AI disproportionately **benefits strong verifiers** while disadvantaging weak verifiers, widening quality differences

AI amplifies verification ability: high- α workers gain compliance, low- α workers risk compliance loss.

Takeaways, Summary, and Future Work

Worker side

- Work becomes a **delegation–verification** choice
Workers choose not only whether to delegate, but how much to **verify**
- **Verification** becomes the sorting ability
High verification reliability enable upgrade; weak verification causes loss with rationally over-delegate

Institution side

- AI is **a structural filter**, not just a productivity tool
AI amplifies differences in verification ability: strong verifiers benefit, weak verifiers may be downgraded
- **Better AI is not always better**
Higher AI capability may trigger rational over-delegation among low- α workers
- **Oversight** must be designed
Use verification protocols, correction rules, and rewards for **substantive checking**

Summary

- Built **delegation–verification** model for AI-assisted institutional work
- Defined worker quality as **institutional utility** under the worker-optimal workflow
- Characterized workflow regimes and **phase transitions**: manual, pure delegation, verified delegation
- Identified **verification amplification**: compliance gain/loss and **rational over-delegation**
- Analyzed type-specific interventions with **heterogeneous** and **non-monotonic** effects
- Tested robustness across action, belief, task, workflow, output, and regularity variants

Limitations and future work

- Extend beyond the **static single-worker** setting: dynamic learning, deskilling, and multi-worker collaboration
- Move from structural analysis to **empirical estimation**: quantify compliance gain/loss in real domains
- From oversight to **governance**: workflow observability, strategic behavior, responsibility, and fairness

Thank you!

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